S. Harsha

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Professional Summary

Software Engineer proficient in Python, FastAPI, and relational database systems.Interested in designing scalable RESTful services and deploying end-to-end full-stack web applications to cloud with clean code architecture.

Projects

Centralized College Counseling & Seat Allocation Platform | GitHub | Live Demo **2026**

- Engineered an automated seat allotment system processing student ranks and preferences utilizing the **Gale-Shapley Deferred Acceptance Algorithm** for deterministic matching.
- Designed a modular, layered backend architecture separating responsibilities into controllers, services, repositories, schemas, and ORM database models.
- Developed secure cookie-based **JWT authentication** enforcing **HttpOnly**, **Secure**, and **SameSite=None** flags alongside **pwdlib** password hashing.
- Built robust backend RESTful APIs using **FastAPI** with **Pydantic** validation, persisting student registries and seat records on **Supabase (PostgreSQL)**.
- Constructed single-page client dashboards using **React.js** with **Axios** credentialed requests for real-time choice filling and status tracking.
- Configured asynchronous background dispatching via the **SMTP protocol** to render and email PDF allotment receipts to authenticated students.
- **Technologies**: FastAPI, React.js, Supabase, Render, SMTP

Smart Expense Tracker & Analytics Platform | GitHub | Live Demo **2026**

- Architected backend micro-services using **FastAPI** following an MVC-inspired layered design pattern to log, aggregate, and calculate net user financial balances.
- Implemented relational database structures using an Object-Relational Mapper backed by **Supabase (PostgreSQL)** with explicit foreign-key entity mappings.
- Engineered session security featuring **JWT access tokens** stored in **HttpOnly/Secure cookies** and protected cross-origin resource sharing (**CORS**).
- Developed custom reusable frontend architecture, including a generic **React** data table with debounced search, drawer inputs, and dynamic UI states.
- Designed automated reporting pipelines utilizing **Matplotlib** to synthesize category-wise expense distributions into dynamic visual charts and PDF reports.
- Configured non-blocking background task execution via **SMTP** to dispatch date-filtered PDF statements directly to registered user email addresses.
- **Technologies**: FastAPI, React.js, Supabase, Render, SMTP

Technical Skills

Languages & Front-end: Python, JavaScript, React.js, HTML5, CSS
Backend Frameworks & Protocols: FastAPI, RESTful APIs, Pydantic, SMTP
Databases: PostgreSQL, Supabase
Developer Tools: Render (Deployment), Git, GitHub, VS Code, Vercel

Education

Jyothy Institute of Technology (JIT) **2021 - 2025**
Bachelor of Engineering (B.E.) in Computer Science and Engineering – ***CGPA: 8.54*** *Bangalore, India*